

CM0832 - MT5001 Elements of Set Theory
The Theory of Zermelo–Fraenkel with Choice

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Preliminaries

Textbook

Karel Hrbacek and Thomas Jech ([1978] 1999). Introduction to Set Theory.

Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

Preliminaries

Acronyms

ZFC Zermelo–Fraenkel set theory with Choice

FOL First-Order Logic

Outline

First-Order Logic

The ZFC Theory

Properties

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First-Order Logic

Theorem

The identity (equality) relation is an equivalence relation.

First-Order Logic

Theorem

The identity (equality) relation is an equivalence relation.

That is, for all X, Y, Z ,

- (a) $X \doteq X$ (reflexivity),
- (b) if $X \doteq Y$, then $Y \doteq X$ (symmetry),
- (c) if $X \doteq Y$ and $Y \doteq Z$, then $X \doteq Z$ (transitivity).

First-Order Logic

Definition

A **proposition** (or **statement**) is a sentence that can be assigned a truth value of **true** or **false**.

First-Order Logic

Definitions

A **propositional function** (or **property**) is a sentence containing one or more variables (parameters) that becomes a proposition when specific values are assigned to its variables.

First-Order Logic

Definitions

A **propositional function** (or **property**) is a sentence containing one or more variables (parameters) that becomes a proposition when specific values are assigned to its variables.

The **domain** of a propositional function is the **domain of the discourse**.

The **codomain** of a propositional function is $\{\text{true}, \text{false}\}$.

First-Order Logic

Theorem

The identity (equality) relation is substitutive.

First-Order Logic

Theorem

The identity (equality) relation is substitutive.

That is, let $\mathbf{P}(X)$ be a propositional function. For all X, Y , if $X \doteq Y$ and $\mathbf{P}(X)$, then $\mathbf{P}(Y)$.

Outline

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Pure Sets

Definition

A **pure set** is a set such that its members are also sets.

Pure Sets

Definition

A **pure set** is a set such that its members are also sets.

Description

The ZFC theory is a first-order theory of pure sets.

First-Order Language of ZFC

Description

In our informal but axiomatic presentation of ZFC we have two primitive (undefined) concepts: **set** and **membership relation**.

First-Order Language of ZFC

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In our informal but axiomatic presentation of ZFC we have two primitive (undefined) concepts: **set** and **membership relation**.

The (binary) membership relation is denoted $\in$.

First-Order Language of ZFC

Notation

$$X \neq Y \stackrel{\text{def}}{=} \neg(X = Y),$$

$$X \notin Y \stackrel{\text{def}}{=} \neg(X \in Y).$$

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Properties

Example

We define the following unary property:

$\mathbf{P}(X)$: “There exists no $Y \in X$ ”.

Properties

Example

We define the following unary property:

$\mathbf{P}(X)$: “There exists no $Y \in X$ ”.

Since that we can prove that there is a **unique** set X with the property $\mathbf{P}(X)$ we can **add** a new **constant** symbol $\emptyset$ denoting the empty set.

Properties

Example

We define the following binary property:

$P(X, Y)$: “Every element of X is an element of Y ”.

Properties

Example

We define the following binary property:

$\mathbf{P}(X, Y)$: “Every element of X is an element of Y ”.

Using the above property we can **add** a new binary **relation** symbol $\subseteq$ denoting that X is a subset of Y :

$$X \subseteq Y \stackrel{\text{def}}{=} \mathbf{P}(X, Y).$$

Properties

Example

We define the following ternary property:

$Q(X, Y, Z)$: “For every U , $U \in Z$ if and only if $U \in X$ and $U \in Y$ ”.

Properties

Example

We define the following ternary property:

$Q(X, Y, Z)$: “For every U , $U \in Z$ if and only if $U \in X$ and $U \in Y$ ”.

Since that we can prove that for every X and Y there is a **unique** Z such that $Q(X, Y, Z)$, we can **add** a new binary **function** symbol $\cap$, denoting the intersection of X and Y :

$$X \cap Y \stackrel{\text{def}}{=} \text{the unique } Z \text{ such that } Q(X, Y, Z).$$

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References

Karel Hrbacek and Thomas Jech [1978] (1999). Introduction to Set Theory. Third Edition, Revised and Expanded. Marcel Dekker (cit. on p. 2).